SERIES: A

QP CODE: 24900085

SERIES: A **Reg No:.....**

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Multi-Disciplinary Course - MG1MDCECT102 - DATA ANALYTICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hours Maximum Marks: 35

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A
Multiple Choice Questions
Answer any **35** questions
Each question carries **one** mark

1 Why are surveys often used in data analytics to collect information from large populations? [U] [2]

a) They reduce the need for analysis
b) They can efficiently collect responses from a large number of participants
c) They are a slow data collection method
d) They only provide qualitative data

2 Which of the following is a qualitative data collection method? [K] [2]

a) Survey
b) Interview
c) Questionnaire
d) Data mining

3 What is the primary purpose of a line chart? [U] [2]

a) To show parts of a whole
b) To display trends over time

c) To compare categorical data
d) To represent frequency distributions

4 Which of the following best describes the x-axis in a typical line chart? [U] [2]
a) It usually represents categorical data.
b) It typically represents time or ordered data.
c) It is always quantitative.
d) It is not used in line charts.

5 What is a challenge commonly faced when analyzing text data? [U] [1]
a) High variability in text format
b) Difficulty in visual representation
c) Lack of numerical values
d) All of the above

6 Which of the following is an example of audio data in multimedia? [U] [1]
a) A song saved as an MP3 file
b) An image of a landscape
c) A PDF document
d) A spreadsheet of sales numbers

7 Which of the following platforms is commonly used for storing and sharing large multimedia files? [U] [1]
a) WordPress
b) YouTube
c) Google Docs
d) Excel

8 In which of the following scenarios is text data most likely to be useful? [A] [1]
a) Measuring the height and weight of a population
b) Calculating the total sales revenue of a company
c) Understanding customer reviews on a product or service
d) Predicting future sales based on historical data

9 What is the primary purpose of data analytics? [U] [1]
a) Storing data
b) Predicting future trends
c) Visualizing data only
d) Collecting data

10 In a study about workplace culture, researchers decide to use interviews, open-ended surveys, and group discussions. Which of these methods primarily collects qualitative data? [A] [1]
a) Structured surveys with numerical ratings
b) Open-ended surveys and interviews
c) Time-tracking apps
d) Structured questionnaires with fixed options

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11 Which type of analytics helps in identifying the reasons behind past outcomes? [U] [1]

a) Predictive analytics
b) Descriptive analytics
c) Diagnostic analytics
d) Prescriptive analytics

12 Which of the following tools is commonly used for data visualization in data analytics? [K] [1]

a) Python
b) Excel
c) Tableau
d) Microsoft Word

13 Name the type of research that commonly uses secondary data. [K] [1]

a) Experimental research
b) Observational research
c) Desk research
d) Field research

14 Identify the form secondary data often takes. [K] [1]

a) Raw, unprocessed data
b) Summarized and analyzed reports
c) Data collected through personal interviews
d) Observational notes

15 Select the correct approach to using secondary data in data analytics. [U] [1]

a) Ignoring the original purpose of the data
b) Using it as the sole source of data without any verification
c) Evaluating the data's relevance and accuracy for the current study
d) Treating secondary data as always more reliable than primary data

16 Which of the following is an example of unstructured data? [K] [1]

a) Sales report in a spreadsheet
b) Social media posts
c) Stock price data in a database
d) Customer phone numbers

17 Which programming language is widely used for data analysis and visualization? [K] [1]

a) Java
b) C++
c) Python
d) Ruby

18 Select the software tool commonly used to process raw data. [K] [1]

a) Microsoft Word
b) Excel
c) Photoshop
d) PowerPoint

19 Select the correct description of how structured data is organized. [U] [1]

a) Data is randomly placed in documents
b) Data is only visualized, not stored
c) Data is organized in rows and columns
d) Data is free-form without any fields

20 Select the type of data that can be easily stored as structured data. [U] [1]
a) Social media conversations
b) Audio recordings
c) Customer transaction records
d) Video clips

21 Select the correct characteristic of discrete data. [K] [1]
a) It can take any value within a range
b) It is measured in intervals
c) It takes whole, countable numbers
d) It has infinite decimal places

22 Identify a field where discrete data is frequently used. [U] [1]
a) Physics experiments
b) Demographic studies (e.g., number of people)
c) Fluid dynamics
d) Weather forecasting

23 Select the best method for collecting continuous data. [U] [1]
a) By measuring with a device like a ruler or thermometer
b) By counting distinct items in a set
c) By asking yes/no questions
d) By grouping data into categories

24 Name the disadvantage of primary data collection. [K] [1]
a) It is limited in scope
b) It is difficult to access
c) It is time-consuming and expensive
d) It is unreliable and inaccurate

25 Identify a common visualization tool used in big data analytics. [K] [1]
a) Microsoft Excel
b) Tableau
c) Notepad
d) Adobe Acrobat

26 Name a challenge associated with big data in data analytics. [U] [1]
a) Simplicity in data collection
b) Lack of available data
c) Data privacy and security concerns
d) Easy visualization

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27 How is navigation data typically used in descriptive analytics? [U] [1]

a) To predict customer needs based on location data
b) To develop long-term strategic plans
c) To forecast future movements of vehicles
d) To understand historical patterns in traffic or route choices

28 Choose the key component of predictive analysis. [K] [1]

a) Machine learning models and algorithms
b) Email marketing campaigns
c) Data summarization
d) Data entry

29 Quantitative data can be categorized into which of the following types? [K] [1]

a) Nominal and Ordinal
b) Subjective and Objective
c) Qualitative and Categorical
d) Discrete and Continuous

30 What can organizations achieve by analyzing operational data? [U] [1]

a) Increased inefficiencies
b) Reduced productivity
c) Optimized processes and cost savings
d) Ignored operational challenges

31 If two rows in your dataset have the same values for all columns except for one, how can you handle this redundancy? [A] [2]

a) Drop both rows
b) Combine the rows into one and decide how to handle the differing value based on the context (e.g., average the value, use the latest entry)
c) Keep both rows to avoid losing data
d) Remove all rows containing missing values

32 You are tasked with understanding the motivations of employees in your company. Which data collection method would be most appropriate to apply for gathering in-depth personal insights? [A] [2]

a) Focus group
b) Survey
c) Interview
d) Data mining

33 What is the best practice for labeling the lines in a line chart? [U] [2]

a) Use generic names
b) Use abbreviations
c) Label each line clearly with a
d) Do not label them at all

legend

34 What is the difference between "Save" and "Save As"? [A] [2]

a) "Save" creates a backup copy.
b) "Save As" allows you to rename the file or change its format.
c) "Save" always prompts for a location
d) There is no difference; they are the same

35 Which of the following formats can you NOT import into Excel? [U] [2]

a) .xml
b) .xlsx
c) .txt
d) .jpg

36 Why is it important to standardize features in high-dimensional datasets? [U] [2]

a) It simplifies data visualization
b) It ensures that each feature has equal importance in distance-based algorithms
c) It makes it easier to impute missing values
d) It removes duplicates from the dataset

37 In which case might data standardization not be necessary? [U] [2]

a) When using algorithms like decision trees or random forests
b) When using distance-based algorithms
c) When the data has a wide range of values
d) When performing clustering

38 Which function returns the minimum value from a set of values? [U] [2]

a) MAX
b) MIN
c) AVERAGE
d) SUM

39 What is the purpose of the VALUE function? [U] [2]

a) Converts text to number
b) Formats numbers
c) Checks for text
d) Rounds numbers

40 What does "Power Query" allow you to do with imported data? [U] [2]

a) Edit the data
b) Clean the data
c) Transform the data
d) All of the above

(35 × 1 = 35)

END OF THE QUESTION PAPER

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a) All of the above b) Transform the data

c) Clean the data d) Edit the data

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4	Identify a common visualization tool used in big data analytics. a) Microsoft Excel b) Tableau c) Notepad d) Adobe Acrobat	[K]	[1]
5	If two rows in your dataset have the same values for all columns except for one, how can you handle this redundancy? a) Keep both rows to avoid losing data b) Combine the rows into one and decide how to handle the differing value based on the context (e.g., average the value, use the latest entry) c) Remove all rows containing missing values d) Drop both rows	[A]	[2]
6	Select the type of data that can be easily stored as structured data. a) Social media conversations b) Customer transaction records c) Video clips d) Audio recordings	[U]	[1]
7	What can organizations achieve by analyzing operational data? a) Optimized processes and cost savings b) Ignored operational challenges c) Increased inefficiencies d) Reduced productivity	[U]	[1]
8	Which programming language is widely used for data analysis and visualization? a) Ruby b) Java c) C++ d) Python	[K]	[1]
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b) Collecting data
c) Predicting future trends
d) Storing data

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